

# Module Four:

## Key Recovery

### Grading

*Note: You will need to have a partner for this lab to exchange keys with.*

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**Total points: 30**

☐ **[2.5 points]** Your submission is labeled as “Your submission is labeled as “cyb404\_module04\_lab00\_[G(group#)]\_[nauid]\_[lastname]\_[firstname].pdf.” For example, if I were in Group 1, to submit a file, it would be labeled as cyb404\_module04\_lab00\_G1\_mk959\_keshavarz\_mahsa.pdf.

**FAILURE TO COMPLY WITH THIS STEP CAN LEAD TO A ZERO GRADE.**

☐ **[25.5 points]** You correctly complete the empty text fields based on the values provided to you and to your partner.

☐ **[2 points]** Provide the name of your partner and their email address: Nasya Valenzuela

### Part 1: Key Recovery

1. The lab instructor will supply each of you with a different three-digit key value. Enter this value in cell B6. Record this value below:

623

2. If they're not already there, enter the values 11, 13 and 16 in cells B14, B15 and B16, respectively. These values are your *moduli*.<sup>1</sup> (It is not absolutely necessary to use the three moduli above. You could choose any three numbers between the cube root of 1000 – 10 – and the square root of 1000 – approximately 31 – such that no pair of moduli has any common factor larger than 1. For example, you could choose 14, 15 and 23.)

Record your three moduli below:

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<sup>1</sup> Moduli is the plural of *modulus*, a term you encountered in the first encryption exercise. These moduli are not really the same as the modulus in the RSA public-key encryption method. However, they are both applications of the same basic mathematical concept.

11            13            16

3. The spreadsheet will split your key into three pieces by dividing by each of the moduli and taking the remainder. The three pieces of your key will appear in cells C14, C15 and C16. Record the values of these pieces below:

7            12            15

4. In reality, one would ask three different parties to each store one piece. Instead, write down your three moduli and the corresponding three pieces in an email or Slack direct message and exchange pieces with your partner student.
5. Click on the tab labeled **Key Recovery**. This reveals a worksheet for recovering a key from its three parts.
6. Enter the *other student's* moduli in cells B7, B8 and B9. Enter her/his corresponding key pieces in cells C7, C8 and C9. Record all six of these values below:

| <u>Modulus</u> | <u>Corresponding piece</u> |
|----------------|----------------------------|
| 11             | 8                          |
| 13             | 11                         |
| 16             | 13                         |

The key recovery calculation takes place in cells C7:E9 and cell E11. It works as follows:

- In each row of column C, compute the products of the moduli in the other two rows. For example, if the moduli in column B are 11, 13 and 16, the corresponding cells in column C will contain  $13 \times 16 = 208$ ,  $11 \times 16 = 176$ , and  $11 \times 13 = 143$ , respectively
- Each row in column D contains the *inverse* of the value in column C with respect to the modulus in column B. To see this, enter the following formula in cell F7:

**= MOD (C7\*D7,A7)**

**MOD** is a built-in function that divides its first argument by its second and takes the remainder. What value appears in cell when you enter this formula?

1

Select the range F7:F9 and fill down. What values appear in cells F8 and F9?

F8: 1            F9: 1

All of these values should be 1. This illustrates the concept of an *inverse*.

- The “magic numbers” in column E are just the product of the inverse and the other moduli. Thus, the magic number in cell E7 is just  $C7 \times D7$ . Note that this is just the product of the number in cell C7 and its inverse.
- To recover the key, multiply each piece by the corresponding “magic number.” Divide the result by the product of the moduli and take the remainder. Because of the special way we have constructed

the magic numbers, this calculation produces the original key.

7. The recovered key value should appear in cell E11. Record this value below. Check with the other student to see that you have correctly recovered their key.

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